

Ferralic Planosols:

Ferralic Planosols are tropical soils with a light-colored, bleached, sandy or loamy surface horizon that shows signs of periodic water stagnation, abruptly overlying a dense, slowly permeable clayey subsoil.

They are named “Ferralic” due to high iron oxide content in the subsoil, giving the soil a reddish-brown color and low base saturation, which reduces fertility compared to eutric soils. Planosols typically form on flat or slightly undulating terrain in seasonally wet areas and may exhibit stagnic properties in the surface horizon during wet periods.

Drainage:

- **Wet season:** Poor to imperfect drainage in surface horizon; water may stagnate above dense subsoil.
- **Dry season:** Surface may harden, while subsoil remains dense and slowly permeable.
- **Landscape:** Flat to gently undulating terrain; frequently in low-lying areas or basin margins.

Depth:

- Surface horizon: 15–30 cm, light-colored and coarse-textured.
- Subsoil: Dense clay layer (>50 cm), with abrupt textural change from topsoil; rooting may be restricted by slow permeability.

Workability:

- **When wet:** Topsoil becomes sticky and plastic; difficult to handle if waterlogged.
- **When dry:** Upper horizons harden; ploughing is difficult; subsoil remains dense.

Nutrient Richness & Cation Exchange Capacity (CEC):

- Low to moderate CEC due to clay destruction and leaching.(less than 16 cmol(+) kg-1 clay within 100 cm from the soil surface.
- Low base saturation and low natural fertility.
- High iron oxide content in subsoil; phosphorus availability is limited due to fixation.
- Low organic matter in surface horizons unless amended.

pH:

- Acidic (pH 4.5–5.5), with low buffering capacity.

Management:

- Improve drainage through surface contouring, raised beds, or ridges to prevent waterlogging.
- Incorporate organic matter (manure, compost) to improve fertility, soil structure, and water retention.
- Apply phosphorus fertilizers carefully (rock phosphate and superphosphate) due to low availability.
- Avoid heavy machinery in wet periods to prevent compaction.
- Suitable crops include wetland rice in low-lying areas, dryland fodder crops, or extensive grazing.